

The Limitations of SMOTE and Class Imbalance in Low-Sample Radiomics: A Cautionary Case Study in Renal Mass Differentiation

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Abstract— Three-dimensional radiomics is widely proposed as a noninvasive tool to differentiate malignant Renal Cell Carcinoma (RCC) from benign Angiomyolipoma (AML), a distinction frequently ambiguous on standard CT imaging due to fat-poor AML variants. However, current literature heavily relies on the Synthetic Minority Over-sampling Technique (SMOTE) to force pipeline stability without sufficient validation processes to verify its legitimacy. This study serves as a critical methodological intervention, evaluating whether feature-space oversampling can anchor a stable decision boundary when pipelines are bound to small, severely imbalanced cohorts. Using arterial phase CT data (N = 69; 58 RCC, 11 AML) to extract 1,182 volumetric features, we implement a dual-state comparison evaluating Support Vector Classifier (SVC) pipelines explicitly in "with-SMOTE" versus "without-SMOTE" configurations. While isolated data partitions yielded deceptive single-fold peaks up to a maximum AUC of 0.94, our multi-fold evaluation exposed a complete performance collapse to random chance. Tabular SMOTE yielded no genuine gains over the unhandled baseline (AUC = 0.49 ± 0.19 versus 0.51 ± 0.20), demonstrating that linear interpolation among scarce minority vectors (N < 15) within high dimensional space (D > 1000) merely generates unphysical mathematical noise. These proofs evidence that raw radiomics features lack a separable biological signal for this task within a single-institution, single-phase scope. To prevent false optimism, future radiomics validation must prioritize image-level data augmentation on source volumes, rather than abstract feature-space shortcuts, in order to preserve genuine anatomical structures and the physical information generated by X-ray interaction within biological tissues.

Keywords — radiomics, renal cell carcinoma, angiomyolipoma, computed tomography (CT), class imbalance, SMOTE, nested cross-validation, machine learning, support vector machines, Decision boundary, Data augmentation

I. INTRODUCTION

In contemporary renal oncology, the accurate differentiation of renal masses is the cornerstone for determining appropriate medical or surgical pathways. The primary classification boundary lies between malignant Renal Cell Carcinoma (RCC) and benign Angiomyolipoma (AML). RCC is inherently invasive and metastatic, demanding aggressive interventions such as partial or radical nephrectomy, targeted therapies, or immunotherapies. Conversely, AML is a benign mesenchymal tumor that typically follows a conservative "watch and wait" surveillance protocol, rendering surgical intervention clinically unnecessary [1].

A severe clinical dilemma arises from the visual overlap between these two entities on standard computed tomography (CT) scans. Specifically, fat poor AMLs (fp AML) lack the

macroscopic fat signatures typically used by radiologists for definitive diagnosis, causing them to look identical to RCCs during standard visual evaluations. Because human vision alone cannot reliably separate these subvisual texture distributions, a high rate of unnecessary nephrectomies is performed on patients with completely benign disease [2].

Radiomics has been proposed as a means of moving "beyond the Hounsfield Unit," converting standard medical images into minable, high-dimensional feature spaces capable of revealing sub-visual biomarkers correlated with underlying tissue biology [3–7]. However, the rare and binary nature of pathologies such as AML means that radiomics cohorts are frequently small and severely class imbalanced. These are precisely the conditions under which both feature extraction and resampling techniques are least theoretically justified. The vast majority of published radiomics studies report performance using k fold cross validation or, in many cases, a single train test split, often without disclosing the variance across folds [13]. When a dataset is small and one class is rare, this practice creates a substantial risk: a single "lucky" partition of the data can produce an impressively high headline metric that does not represent the model's true, generalizable behavior.

This study uses the clinically important RCC-versus-AML differentiation task as a controlled case study to examine this broader methodological problem. Rather than presenting a single performance figure, we deliberately foreground the full distribution of outcomes obtained under rigorous nested cross validation, and we use this distribution to investigate why a standard, widely adopted oversampling technique (SMOTE) fails to produce a stable decision boundary when only a handful of original minority class examples are available. Our goal is not to dismiss radiomics as a diagnostic tool, but to demonstrate, transparently and quantitatively, the conditions under which its reported performance should be interpreted with caution, and to propose a more principled path, namely image level data augmentation toward genuinely stable models in low sample, high dimensional medical imaging tasks.

II. MATERIALS AND METHODS

A. Collection Data And Cohort Selection:

This study utilizes high-fidelity imaging data from the UCSF RMaC (University of California San Francisco 3D Multi Phase Renal Mass CT Dataset with Tumor Segmentations) open-source repository [8]. To maintain the integrity of the original Hounsfield Units (HU) and avoid

artifacts from manual manipulation, the data was utilized in its current high-fidelity state.

Cohort selection was handled via a deterministic filtering pipeline based on the dataset's provided metadata:

- **Phase Selection:** The entire cohort was filtered to isolate contrast enhanced arterial phase images.
- **Pathology Filtering:** Images were stratified by confirmed histopathology to isolate Carcinoma and Angiomyolipoma.
- **Final Subcohort:** Metadata filtering yielded a final operational subcohort of 69 cases, comprising 11 angiomyolipoma cases and 58 carcinoma cases.

B. Radiomics Pipeline and Mathematical Foundations

The workflow extracts standardized 3D volumetric radiomics features across the entire segmented tumor mass, avoiding the loss of spatial data inherent to 2D single slice boundaries. A total of 1182 features were extracted using the Python based PyRadiomics toolkit. The computational implementation is built on a structured mathematical backbone where advanced matrices analyze spatial voxel interactions [9]:

- **Image Standardization and Device Normalization:**

Raw slice thicknesses in the UCSF RMaC cohort ranged from 0.625 mm to 10 mm, and in-plane spatial resolutions varied between 0.42 mm and 1.36 mm [8]. To overcome variability across different acquisition devices and ensure texture comparability, all volumes were resampled to a 1 mm $\times$ 1 mm $\times$ 1 mm isotropic spatial resolution using the Python (version 3.12) utilizing the SimpleITK library. Furthermore, to compensate for variations in Hounsfield Unit (HU) distributions across different scanners, intensity values were scaled using Z score normalization [10].

- **Mask Geometry and Phase Alignment:**

To keep some of the images usable during the contrast-enhanced arterial phase, a registration tier (a multi stage spatial alignment process) was implemented to resolve mismatches between image and mask geometry. For cases where dimensions varied, the tumor masks served as the fixed template, and the arterial phase were resized and coregistered using spline interpolation to preserve regional information and ensure the 3D volume accurately encapsulated the segmented tumor mass.

- **Feature Categorization And Algorithmic Discovery:**

The extracted feature space encompasses baseline 3D shape descriptors capturing macroscopic growth behaviors such as sphericity, flatness, and elongation to differentiate the well defined boundaries of benign angiomyolipomas (AML) from the chaotic, asymmetric margins of malignant renal cell carcinomas (RCC) alongside advanced intensity and microstructural metrics. These metrics were computed both on the baseline images (including interpolation diagnostics) and across mathematically filtered spaces using Laplacian of Gaussian (LoG sigma) filters at varying spatial scales and

multi-directional wavelet decompositions. Within these spaces, first order statistics quantified global gray-level distributions (e.g., mean, variance, skewness, and entropy) to evaluate baseline tumor density, while advanced second and higher order textural matrices including the Gray Level Cooccurrence Matrix (GLCM) and the Gray Level Run Length Matrix (GLRLM), calculated complex voxel-pair joint probabilities and consecutive intensity runs to mathematically chart structural tissue heterogeneity [9,11].

C. Downstream Machine Learning Designs:

To address the severe class imbalance (~84% carcinoma vs. ~16% angiomyolipoma) resulting from the limited number of angiomyolipoma cases, a rigorous nested cross-validation framework is implemented. Because this limited dataset is not intended for immediate clinical deployment, the primary objective of this nested framework is to reliably evaluate and select the optimal machine learning algorithm to align with data profile, best features to overcome overfitting issues and balance computational resources for differentiating between these two distinct tumor types.

Within each outer loop, test and training data were strictly isolated to prevent data leakage. Within the inner loops, a sequential machine learning pipeline, comprising standard scaling, ANOVA based feature selection (SelectKBest), Grid Search hyperparameter optimization, and classification, was executed independently across a variety of algorithms (Random Forest, XGBoost, K-Nearest Neighbors (KNN), Logistic Regression, and Support Vector Machines (SVM)), both with and without SMOTE oversampling (Table I). While tree-based ensembles like XGBoost and Random Forest occasionally yielded comparable or marginally higher localized performance metrics, they introduced significant computational overhead. Random Forest models, in particular, frequently caused execution timeouts and ran into operational caps under standard hardware profiles with restricted runtime allocations due to the overhead of building large ensembles of high-dimensional trees. Consequently, the SVM was retained as the primary focus of this analysis. It offers a distinct resource consumption advantage by offering deterministic, highly efficient training via convex optimization (Table II), while structurally remaining the most theoretically suitable candidate for boundary focused classification in overlapping distributions [12,13]. Most importantly, its clear geometric dependency on support vectors allows its outer-fold behavior to most transparently expose the exact boundary instability introduced by feature space oversampling.

Crucially, this workflow is identical in structure to what is reported as a success in much of the published radiomics literature: a multifold or single split, validation is run, a representative metric is computed, and that metric is reported as evidence of model validity. The purpose of the following sections is to show what this standard workflow conceals when it is not accompanied by full disclosure of per fold variability.

TABLE I. CLASSIFIER PERFORMANCE TRADE OFFS : COMPARATIVE ANALYSIS OF SINGLE-FOLD PEAKS, NESTED CROSS-VALIDATION VOLATILITY (MEAN $\pm$ SD), AND COMPUTATIONAL OVERHEAD WITH AND WITHOUT SMOTE RESAMPLING

Algorithm	Metric	With Smote		Without Smote		Computational Overhead / Resource
		Best Fold Score	Multi-Fold Mean ($\pm$ SD)	Best Fold Score	Multi-Fold Mean ($\pm$ SD)	
SVC	AUC	0.81	0.49 $\pm$ 0.19	0.79	0.51 $\pm$ 0.20	Low (Highly efficient, deterministic training via convex optimization)
	Accuracy	94.1%	86% $\pm$ 6	91%	80% $\pm$ 6	
	F1 Score	0.75	0.49 $\pm$ 0.26	0.66	0.39 $\pm$ 0.18	
XGBoost	AUC	0.90	0.61 $\pm$ 0.14	0.90	0.60 $\pm$ 0.20	Moderate (Gradient boosted trees, moderate memory footprint)
	Accuracy	94.1%	86% $\pm$ 4	85%	85% $\pm$ 0	
	F1 Score	0.75	0.46 $\pm$ 0.15	0	0	
Random Forest	AUC	0.94	0.61 $\pm$ 0.21	0.69	0.57 $\pm$ 0.20	High (Severe resource strain; prone to timeouts under restricted cloud environments)
	Accuracy	94.1%	86% $\pm$ 4	94%	86% $\pm$ 5	
	F1 Score	0.80	0.40 $\pm$ 0.26	0.75	0.38 $\pm$ 0.25	
KNN	AUC	0.81	0.49 $\pm$ 0.25	0.78	0.45 $\pm$ 0.20	Low (Instance-based, but storage-dependent)
	Accuracy	85%	76% $\pm$ 7	88%	85% $\pm$ 1	
	F1 Score	0.54	0.42 $\pm$ 0.12	0.33	0.06 $\pm$ 0.13	
Logistic Regression	AUC	0.52	0.51 $\pm$ 0.15	0.72	0.62 $\pm$ 0.17	Low (Linear baseline)
	Accuracy	88%	72% $\pm$ 13	76%	64% $\pm$ 12	
	F1 Score	0.50	0.44 $\pm$ 0.10	0.55	0.42 $\pm$ 0.09	

III. RESULTS

The proposed radiomics based machine learning framework was evaluated using a nested cross-validation [14] strategy across a variety of classifiers, both with and without SMOTE resampling, to ensure robustness under limited and imbalanced data conditions. Across the five outer folds, all classifiers’ performances varied dramatically depending on how the cohort of 69 cases (and particularly the 11 angiomylipoma cases) was partitioned between training and test sets.

As shown in Table I, while complex tree ensembles achieved impressive localized peak metrics on isolated “lucky” splits (with Random Forest and XGBoost reaching peak outer-fold AUCs of 0.94 and 0.90, respectively), their full multifold distributions revealed a uniform collapse toward random chance. The baseline Support Vector Machine (SVC) mirrored this exact pattern, achieving a single fold peak AUC of 0.81, an accuracy of 94.1%, and an F1 score of 0.75 under

SMOTE configuration. Considered in isolation, this specific split would represent a strong, clinically promising result. However, the full multifold average metrics remained heavily depressed (e.g., SVC mean AUC of 0.49 $\pm$ 0.19, accuracy 86% $\pm$ 6, and F1 score 0.49 $\pm$ 0.26).

Crucially, incorporating the nonresampled baseline revealed that tabular oversampling provided no genuine diagnostic gain over unhandled class imbalance across all models (e.g., SVC mean AUC of 0.51 $\pm$ 0.20, accuracy 86% $\pm$ 6, and F1-Score 0.39 $\pm$ 0.18 without SMOTE). While unhandled imbalance caused structural majority class bias (yielding an absolute zero F1 score across all folds for XGBoost), the introduction of SMOTE merely traded systematic bias for extreme single fold volatility without stabilizing the underlying decision boundary.

Furthermore, Table II reports that the optimal hyperparameter configurations independently selected for SVC within each inner loop were highly unstable across folds, demonstrating that inner loop grid searches converged on

fundamentally different feature counts, balancing weights, and classifier setting depending on the minor variations of the training partitions.

IV. DISCUSSION

The simultaneous evaluation of our pipeline's performance with and without SMOTE reveals a critical methodological reality: tabular oversampling provides no meaningful performance gain over unhandled class imbalance. In both experimental arms, the multifold average performance flatlines at random chance. This lack of improvement indicates that the extracted feature space lacks a stable discriminative signal under raw conditions. However, the true danger lies in the severe single fold volatility introduced by SMOTE, which produced an isolated peak AUC of 0.81 in Fold 2 for SVC. This empirical finding demonstrates how feature space oversampling can easily fabricate the false appearance of a highly successful classification model out of data that is otherwise statistically indistinguishable from noise.

A. Geometric Failure Of SMOTE Under Severe Data Scarcity

This architectural volatility stems directly from the mathematical constraints of high dimensional interpolation. SMOTE synthesizes minority instances by linearly interpolating between an original minority sample and its nearest intra class neighbors [17]. In this study, the interpolation pool was restricted to 11 original AML cases, reducing it to just 8 or 9 vectors within any given training split.

In a feature space of $D = 1182$ dimensions, 8 to 11 points represent extreme geometric sparsity that cannot sample the true distribution of angiomylipoma phenotypes. The resulting synthetic samples are constrained to arbitrary line segments, forming tight *synthetic noise clusters* that reflect the idiosyncrasies of the specific training partition rather than genuine biological variability. As the training pool shifts with each cross validation fold, the synthesized minority region shifts correspondingly, dragging the SVC decision boundary with it.

This aligns with established evidence that SMOTE's low-dimensional benefits do not extend to high dimensional, small sample regimes, where the technique fails to attenuate majority class bias, decreases data variability, and introduces spurious correlation structures [18]. Independent radiomics cohorts similarly confirm that small samples and difficult classification tasks yield highly unstable AUC estimates across different splits [19]. Our findings extend this to feature space oversampling under severe class imbalance: even a margin maximizing classifier like the SVM [12,13] is only as stable as the data defining its support vectors, and 11 original cases cannot reliably anchor that boundary.

B. Baseline Dual-State Empirical Proof

Definitive proof of pipeline failure is only established by analyzing both experimental arms together:

- Baseline Arm (No SMOTE): Mean AUC= 0.51±0.20 → Boundary unresolvable unassisted

- Resampled Arm (With SMOTE): Mean AUC= 0.49±0.19→Oversampling fails to recover signal

Because an industry standard resampling technique completely failed to rescue the baseline and instead merely induced volatile single fold spikes, we draw a definitive two fold conclusion: tabular oversampling is an invalid fix for small minority sets ($N_{\text{minority}} < 15$), and these unaugmented radiomic features lack a robust, separable biological signal for this clinical task [19].

C. The Risk Of Performance Inflation In The Radiomics Literature

A broader implication of these findings concerns reporting practices across the radiomics field. Systematic reviews in medical imaging repeatedly highlight widespread reporting and methodological flaws, characterized by studies reporting single train test splits, selective "favorable folds" or presplit oversampling that causes catastrophic data leakage [13]. Our isolated Fold 2 for SVC result mirrors the inflated headline metrics that populate much of the published small cohort radiomics literature. Only by exposing the full nested cross validation distribution and contrasting the volatile folds does this underlying instability become visible. Transparent distribution reporting, rather than the pursuit of a single optimized metric, must become the standard for data constrained radiomics studies.

D. Shift Toward Image-Level Data Augmentation

These findings motivate a crucial shift in addressing data scarcity. SMOTE and tabular oversampling operate blindly within an abstract, decontextualized mathematical feature space that loses all direct correspondence to anatomical structures once the CT volume is vectorized. Conversely, augmentation performed directly in the image domain preserves spatial anatomy and pixel relationships, anchoring any resulting features in physically and biologically plausible variations:

- Geometric and Intensity Manipulations: Standard 3D random rotations, translations, and scaling of the CT volume, combined with controlled Gaussian noise added to the Hounsfield Unit distribution to simulate multi scanner and protocol variability.
- ROI (Region of Interest) Perturbations: Small modifications to the tumor segmentation boundary (e.g., eroding or dilating the contour by one voxel, or applying contour randomization) [15]. This forces the pipeline to extract features from clinically plausible variations of the same tumor mass, building model stability rather than introducing synthetic statistical noise.

Morphological perturbation chains have already been validated [15] to distinguish robust from nonrobust radiomic features [20] without requiring repeat imaging. Unlike abstract tabular extrapolation, image-level manipulation anchors the pipeline in genuine, 3D anatomical variability.

TABLE II : NESTED CROSS-VALIDATION PERFORMANCE AND OPTIMAL HYPERPARAMETER CONFIGURATIONS FOR SVC

Outer Fold	Metrics AUC ACCURACY F1-SCORE	Smote k_neighbors:	SelectKBest k:	SVC C:	SVC gamma:	SVC kernel:	Confusion Matrix	
							TN	FP
							FN	TP
1	AUC : 0.45 ACC. : 91.1% F1-SCORE : 0.66	6	17	0.5	'auto'	'rbf'	28	1
							2	3
2	AUC : 0.81 ACC. : 94.1% F1-SCORE: 0.75	4	7	50	0.01	'rbf'	29	0
							2	3
3	AUC : 0.56 ACC.: 88.2% F1-SCORE: 0.50	4	9	5	0.1	'rbf'	28	1
							3	2
4	AUC : 0.24 ACC. : 75.7% F1-SCORE : 0.0	5	7	5	0.1	'rbf'	25	3
							5	0
5	AUC : 0.39 ACC: 84.8% F1-SCORE: 0.54	6	17	5	'scale'	'linear'	25	3
							2	3

E. Limitations

Three primary limitations qualify these findings. First, the small cohort size ($N = 69$, 11 AML cases) is a deliberate subject of investigation rather than an incidental constraint; while specific volatility magnitudes vary, the underlying high dimensional interpolation mechanism generalizes to any scarce cohort. Second, this study relies on a single institution dataset and a single imaging phase (arterial); multi center external validation remains necessary before drawing generalizable clinical conclusions. Third, while shifting toward image level augmentation is theoretically well motivated, its capacity to stabilize the SVM decision boundary in this specific task remains a hypothesis for future empirical validation.

V. CONCLUSION AND FUTURE WORK

This study establishes that standard tabular oversampling via SMOTE is structurally incapable of recovering a discriminative signal or anchoring a stable decision boundary when differentiating malignant Renal Cell Carcinoma (RCC) from benign Angiomyolipoma (AML) if the minority class is limited to a handful of high-dimensional vectors. While isolated data partitions yielded deceptive, highly publishable peaks for SVC (AUC of 0.81 to 0.94), rigorous nested cross validation exposed a complete collapse of average performance to chance level. The dual state comparison of "with-SMOTE" versus "without-SMOTE" configurations makes this study methodologically unique compared to the common way of simple pipeline evaluation, serving as a methodological critique addressing widespread reporting

flaws in peer reviewed radiomics literature. Ultimately, these empirical proofs reveal that raw radiomic features lack a separable biological signal in such data scarce regimes, a finding highly valuable to the broader scientific community. This work highlights that medical imaging progress relies not just on publishing successful algorithms, but on rigorously documenting why widespread feature space shortcuts fail in data scarce clinical tasks.

We propose shifting image level data augmentation to break these deceptive metric illusions to ensure radiomics analysis rests on a physically grounded approach. Because this method provides to represent physical interactions between X-ray photons and biological structures and respects spatial reality. Implementing image level manipulation followed by rigorous cross validation represents a methodologically superior hypothesis: it anchors every synthetic sample in genuine 3D tumor anatomy. If future multifold mean performance rises above $AUC = 0.50$ under image augmented frameworks, it will definitively prove that tabular SMOTE was destroying underlying spatial relationships; if it remains at chance level, it will provide conclusive evidence that the radiomic features themselves carry no discriminative power for the clinical task at hand within this current sample scope.

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